



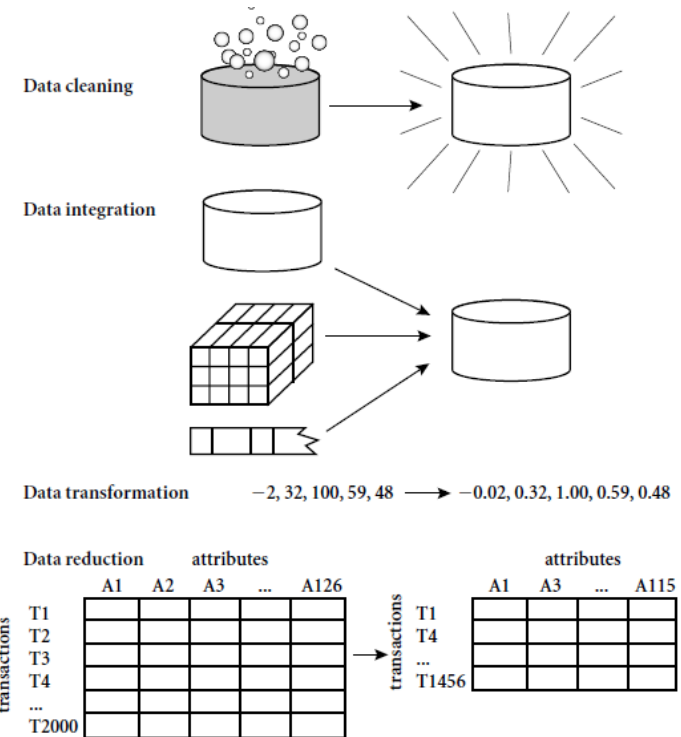
Applied Data Mining

**— Week 4—
Spring 2026**

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Outline

- Data Preprocessing: Definition, Goal
- Why do we preprocess the data?
- Major Tasks in Data Preprocessing
 - Data Cleaning
 - Data Integration
 - Data Reduction
 - Data Transformation



The Pearson Correlation Coefficient

The Pearson correlation coefficient is a statistic that measures the strength and direction of the linear relationship between two continuous variables, ranging from -1 to 1

The diagram shows the formula for the Pearson correlation coefficient r with several annotations:

- x_i are the **individual values** of one **variable** e.g. age (points to x_i in the numerator)
- y_i are the **individual values** of the other **variable** e.g. salary (points to y_i in the numerator)
- $\bar{x}$ and $\bar{y}$ are respectively the **mean values** of the two variables. (points to $\bar{x}$ and $\bar{y}$ in the denominator)
- Where r is the Pearson correlation coefficient, (points to r in the numerator)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The Pearson Correlation Coefficient

The Pearson correlation coefficient is a statistic that measures the strength and direction of the linear relationship between two continuous variables, ranging from -1 to 1

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{Cov(X, Y)}{\sigma(X)\sigma(Y)}$$

Example



Subject	Age x	Glucose Level y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

The Correlation Coefficient is 0.5409

Example



Mean of $X = 247/6 = 41.16$, Mean of $Y = 486/6 = 81$

$$\begin{aligned}\text{COV}(X, Y) = & (43 - 41.16)(99 - 81) + (21 - 41.16)(65 - 81) + (25 - \\ & 41.16)(79 - 81) + (42 - 41.16)(75 - 81) + (57 - \\ & 41.16)(87 - 81) + (59 - 41.16)(81 - 81)\end{aligned}$$

Calculate $\sigma(X)$, and $\sigma(Y)$

$$\begin{aligned}\text{Correlation}(X, Y) &= \text{COV}(X, Y) / \sigma(X) * \sigma(Y) \\ &= 0.5409\end{aligned}$$

Covariance (Numeric Data)

- Covariance is similar to correlation

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$
$$\text{Correlation} = r_{A,B} = \frac{Cov(A, B)}{\sigma_A \sigma_B}$$

- **Positive covariance:** If $Cov_{A,B} > 0$, then A and B both tend to be larger than their expected values.
- **Negative covariance:** If $Cov_{A,B} < 0$ then if A is larger than its expected value, B is likely to be smaller than its expected value.
- **If A, B independent then** $Cov_{A,B} = 0$ but the converse is not true:
 - Some pairs of random variables may have a covariance of 0 but are not independent. Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence

Covariance: An Example

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

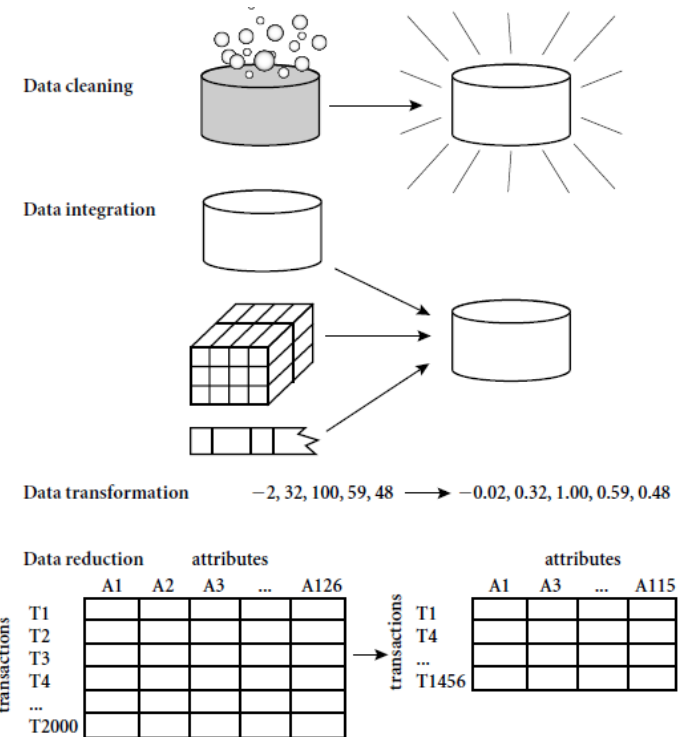
- It can be simplified in computation as

$$Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

- Suppose two stocks A and B have the following values in one week:
(2, 5), (3, 8), (5, 10), (4, 11), (6, 14).
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
 - $Mean(A) = (2 + 3 + 5 + 4 + 6) / 5 = 20/5 = 4$
 - $Mean(B) = (5 + 8 + 10 + 11 + 14) / 5 = 48/5 = 9.6$
 - $Cov(A, B) = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$
- Thus, A and B rise together since $Cov(A, B) > 0$.

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Data Reduction Strategies

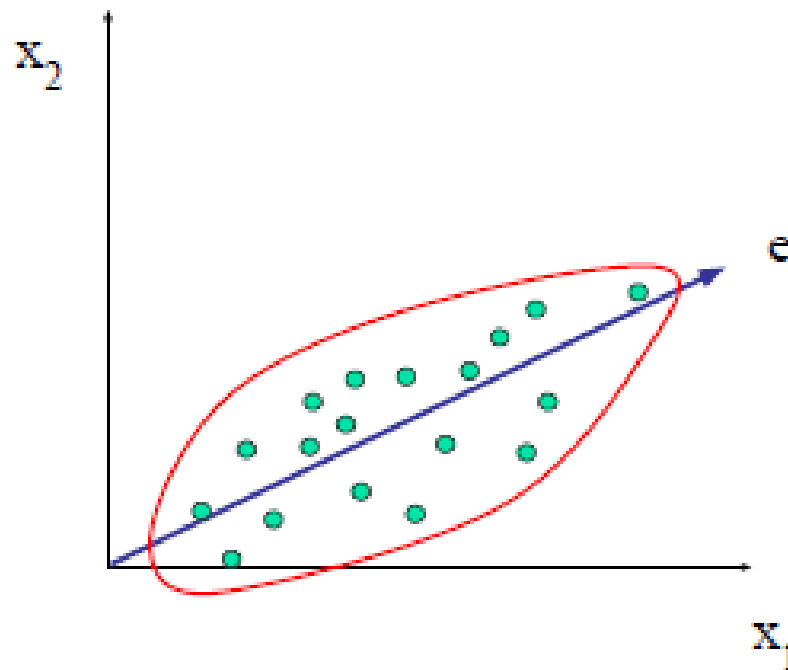
- **Data reduction:** Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results.
- Why data reduction? — A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.
- Data reduction strategies
 - Dimensionality reduction, e.g., remove unimportant attributes
 - Principal Components Analysis (PCA)
 - Feature subset selection, feature creation
 - Data compression

Data Reduction: Dimensionality Reduction

- **Curse of dimensionality**
 - When dimensionality increases, data becomes increasingly sparse
 - The possible combinations of subspaces will grow exponentially
- **Dimensionality reduction**
 - Avoid the curse of dimensionality
 - Help eliminate irrelevant features and reduce noise
 - Reduce time and space required in data mining
 - Allow easier visualization

Principal Component Analysis (PCA)

- Project or transform the original data onto a smaller space.
- Find a projection that captures the largest amount of variation in data
- The original data are projected onto a much smaller space, resulting in dimensionality reduction. We find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Attribute Subset Selection

- Another way to reduce dimensionality of data
- Redundant attributes
 - Duplicate much or all of the information contained in one or more other attributes
 - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
 - Contain no information that is useful for the data mining task at hand
 - E.g., students' ID is often irrelevant to the task of predicting students' GPA

Heuristic Search in Attribute Selection

- How can we find a 'good' subset of the original attributes?
- There are 2^d possible attribute combinations of d attributes
- The "best" (and "worst") attributes are typically determined using tests of statistical significance, which assume that the attributes are independent of one another.
- Many other attribute evaluation measures can be used such as the *information gain measure* used in building decision trees for classification.

Heuristic Search in Attribute Selection

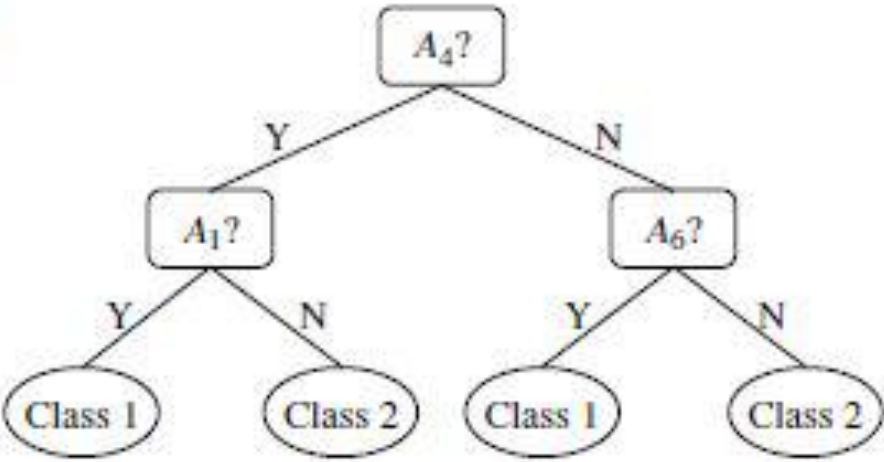
- Best step-wise feature selection (forward selection):
 - Starts with an empty set of attributes as the reduced set.
 - The best of the original attributes is determined and added to the reduced set.
 - At each subsequent iteration, the best of the remaining original attributes is added to the set.
- Step-wise attribute elimination (backward elimination):
 - The procedure starts with the full set of attributes. At each step, it removes the worst attribute remaining in the set.

Heuristic Search in Attribute Selection

- Combination of FS and BE.
- At each step, the procedure selects the best attribute and removes the worst from the remaining attributes.
- **Decision tree induction:**
 - Decision tree induction constructs a flowchart like structure where each internal (non leaf) node denotes a test on an attribute.
 - Each branch corresponds to an outcome of the test, and each external node denotes a class prediction.
 - At each node, the algorithm chooses the “best” attribute to partition the data into individual classes.

Attribute Subset Selection

Heuristic Search in Attribute Selection

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  <pre> graph TD A4["A4?"] -- Y --> A1["A1?"] A4 -- N --> A6["A6?"] A1 -- Y --> C1_1((Class 1)) A1 -- N --> C2_1((Class 2)) A6 -- Y --> C1_2((Class 1)) A6 -- N --> C2_2((Class 2)) </pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

Data Reduction: Data Compression

- String compression
 - There are extensive theories and well-tuned algorithms
 - Typically lossless, but only limited manipulation is possible
- Audio/video compression
 - Typically lossy compression, with progressive refinement
 - Sometimes small fragments of signal can be reconstructed without reconstructing the whole

Data Compression

